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Lab 1: Implementation of Source code management using git commands through github

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OBJECTIVE:

Implementation of Source Code Management Using Git Commands Through GitHub.

AIM

To understand and implement Source Code Management (SCM) using Git and GitHub for tracking changes, managing versions, and collaborating on software projects.

THEORY

Source Code Management (SCM) is the process of tracking and controlling changes in software source code. SCM helps developers work collaboratively, maintain version history, and recover previous versions when necessary.

Git is a distributed version control system that enables developers to record changes to files and coordinate work among multiple developers.

GitHub is a cloud-based hosting platform that provides Git repository management along with collaboration tools such as pull requests, issue tracking, and project management.

The combination of Git and GitHub is widely used in Agile Software Development because it facilitates teamwork, continuous integration, version tracking, and code review.

OBJECTIVES

- Learn the basics of Git.
- Create and manage Git repositories.
- Track project changes using commits.
- Connect local repositories with GitHub.

- Push and pull project updates.
- Understand collaborative software development.

ALGORITHM

1. Install Git on the system.
2. Create a GitHub account.
3. Create a repository on GitHub.
4. Initialize a local Git repository.
5. Add project files.
6. Stage files using Git.
7. Commit changes.
8. Link local repository to GitHub.
9. Push files to GitHub.
10. Verify successful synchronization.

CODE

- Initialize Repository

```
git init
```

- Configure User Information

```
git config --global user.name "your name"
git config --global user.email "your@email.com"
```

- Check Repository Status

```
git status
```

- Add Files

```
git add .
```

- Commit Changes

```
git commit -m "Initial Commit"
```

- Connect Repository to GitHub

```
git remote add origin https://github.com/username/repository.git
```

- Push to GitHub

```
git branch -M main
git push -u origin main
```

- Pull Latest Changes

```
git pull origin main
```

OUTPUT

The Git repository was successfully initialized and connected to GitHub.

- Repository created successfully.
- Files tracked using Git.
- Changes committed successfully.
- Remote repository linked.
- Project pushed to GitHub.

RESULT

The implementation of Source Code Management using Git and GitHub was completed successfully. Version control operations such as repository creation, staging, committing, and synchronization with GitHub were performed successfully.

CONCLUSION

This experiment provided practical knowledge of Source Code Management using Git and GitHub. It demonstrated how version control systems help manage software projects efficiently by maintaining code history, enabling collaboration, and supporting Agile Software Development practices.